Python Fundamentals Notes (Day 1–Day 7)

1. Variables

Variables store values in memory. Example: x = 10

2. Data Types

int, float, str, bool, list, tuple, set, dictionary.

3. Input & Output

input() accepts user input. print() displays output.

4. Operators

Arithmetic: + - * / // % **
Comparison: == != > < >= <=
Logical: and, or, not

5. Conditional Statements

if, elif, else execute code based on conditions.

6. Loops

for loop iterates over sequences.
while loop repeats until condition becomes False.
break exits loop. continue skips current iteration.

7. Strings

Immutable sequence. Concepts: indexing, negative indexing, slicing.
Methods: upper(), lower(), replace(), find(), split(), strip().

8. Lists

Mutable collection using []. Methods: append(), insert(), remove(), pop(), len(). Supports indexing and slicing.

9. Tuples

Immutable collection using (). Supports indexing, slicing, len().

10. Sets

Unique unordered collection using {}. Methods: add(), remove(). Duplicate values are removed automatically.

11. Dictionaries

Store data as key-value pairs. Access using keys. Add/update: dict[key]=value. Remove: pop(key).
Safe access: get(key).

12. Key Differences

List: Mutable []
Tuple: Immutable ()
Set: Unique, unordered
Dictionary: Key-value pairs

13. Interview Tips

Explain what is stored, explain each method, trace the code, justify the output, use technical terms like mutable, immutable, indexing, slicing and key-value pair.

14. Methods Learned

Strings: upper(), lower(), replace(), find(), split(), strip()
Lists: append(), insert(), remove(), pop(), len()
Sets: add(), remove()
Dictionaries: get(), pop(), assignment

15. Revision Checklist

Variables, Data Types, Operators, Conditions, Loops, Strings, Lists, Tuples, Sets, Dictionaries